Hao Lu

PhD Candidate in Robotics

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Google Scholar

School of Computing, Australian National University — Canberra, Australia

Research Profile

Robotics researcher and systems engineer working on reliable robot planning and manipulation, with a focus on planning under uncertainty, sampling-based motion planning, manipulation and rearrangement, task and motion planning, combinatorial optimisation, and decision-making under partial observability. My doctoral research develops planning frameworks with theoretical guarantees while connecting algorithmic structure to practical robotic systems. Prior to the PhD, I spent more than a decade in industrial systems engineering at Freescale/NXP, spanning embedded systems, software/hardware co-design, AI inference, performance optimisation, and customer-facing deployment.

Research Interests

Motion planning; planning under uncertainty; manipulation and rearrangement; task and motion planning; kinodynamic planning; navigation; probabilistic robotics; partially observable decision-making; combinatorial optimisation; nonprehensile manipulation; robot learning and embodied AI; generative/world models for robotics; safety, reliability, and efficient decision-making.

Education

Australian National University (ANU), Canberra, Australia — Feb 2024 – Present
PhD Candidate in Computer Science (Robotics)
Supervisors: Dr Rahul Shome, Prof Hanna Kurniawati, Dr Marco Cheung

Australian National University (ANU), Canberra, Australia — Feb 2021 – Dec 2023
Master of Machine Learning and Computer Vision | GPA: 6.92/7.0
Recipient of the ANU Postgraduate Medal for Academic Excellence **(19 / ~6,000 ANU graduates)**

University of Toronto, Toronto, Canada — Sep 2022 – Apr 2023
Graduate Exchange Student, Department of Computer Science | GPA: 4.0/4.0

Nankai University, Tianjin, China — Sep 2010 – Jul 2013
Master of Engineering in Computer Science / Computer Architecture

Anhui Medical University, Hefei, China — Sep 2005 – Jul 2009
Bachelor's degree (Public Service Management / Medicine program)

Peer-Reviewed Publications

Virtues of Ordered Chaos: Planning with Topple Actions in Tabletop Stack Rearrangement.
H. Lu, R. Shome
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. PDF

Optimal Knock-Pick Planning for Tightly Packed Tabletop Blocks With Parallel Grippers.
H. Lu, R. Shome
International Workshop on the Algorithmic Foundations of Robotics (WAFR), 2026. PDF

Zero-Shot Generative World Priors for Planning and Object Reaching in Partially Observed Rooms.
H. Lu, S. S. Bhattacharjee, D. Campbell, R. Shome
RSS 2026 Workshop on Foundation Models for Robot Planning (FM4RoboPlan), 2026. PDF

Sampling-based Motion Planning for Optimal Probability of Collision under Environment Uncertainty.
H. Lu, H. Kurniawati, R. Shome
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 3138–3145. DOI

Arbitor: A Numerically Accurate Hardware Emulation Tool for DNN Accelerators.
C. Jiang, A. Jayarajan, **H. Lu**, G. Pekhimenko
2023 USENIX Annual Technical Conference (USENIX ATC '23), pp. 519–536, 2023.

Research Experience and Projects

PhD Researcher, Australian National University — Feb 2024 – Present

- Develop algorithms for reliable planning and decision-making under uncertainty, combining sampling-based planning, optimisation, probabilistic modelling, and robotic task/motion reasoning.
- Build end-to-end research systems using ROS2, MoveIt2, OMPL, Isaac Sim, collision checking, mathematical optimisation, and Monte Carlo evaluation; integrate algorithmic prototypes with simulated and real robotic platforms.

Uncertainty-aware manipulation planning — *Oct 2025 – Present*

- Extending probability-of-collision-aware motion planning to manipulation problems with uncertain target-object and obstacle poses.

- Develop physics- and geometry-informed reasoning for grasping and manipulation under uncertainty, with the goal of coupling robust grasp/action feasibility with downstream motion-planning risk.
- Ongoing work targeted for submission to IEEE Robotics and Automation Letters (RA-L).

Kinodynamic planning under environment and control uncertainty *Aug 2026 – Present*

- Developing a sampling-based kinodynamic planning algorithm that extends the uncertainty-aware planning framework to systems without an exact steering function.
- Model uncertainty from both the environment and the control space; sample control commands and durations for forward propagation while evaluating execution outcomes across sampled worlds.
- Target asymptotically optimal planning under uncertainty by combining sparse kinodynamic exploration ideas with scenario-based probability-of-success objectives and pruning/selection mechanisms adapted to uncertain dynamics and environments.

Sampled-World Contingent Task and Motion Planning with Model Repair *Planned*

- Planned framework for online task and motion planning in unfamiliar environments with both uncertain geometry and imperfect symbolic action models.
- Combine evidence-consistent model-repair candidates with geometric world samples into a weighted scenario ensemble; maintain fixed action-model hypotheses within each simulated scenario while updating scenario weights and states from execution observations.
- Couple a configuration-space roadmap for physical motion with a belief graph over weighted scenario states. Observation-dependent branching enables contingent policies that share actions while scenarios remain observationally indistinguishable and diverge when evidence separates them.
- Investigate finite-horizon decision-making under bounded verification budgets, scenario-indexed feasibility checking, and state-qualified caching to trade off immediate task progress, information gathering, computation, and execution risk.

Optimal knock–pick planning for tightly packed tabletop blocks *2025 – 2026*

- Developed an algorithmic formulation for manipulation in tightly packed tabletop scenes where direct picking is obstructed and carefully chosen knock actions can expose graspable objects.
- Derived structured graph abstractions that permit optimal action selection in polynomial time for the studied setting, with applications to packing/unpacking, logistics, and dense warehouse manipulation.
- Published at WAFR 2026.

Planning with toppling actions for tabletop stack rearrangement *Feb 2025 – 2026*

- Introduced toppling as a purposeful nonprehensile action for rearranging dense tabletop stacks rather than treating unstable interactions purely as failures to avoid.
- Developed a graph-based toppling abstraction and a partially grounded task-and-motion planning framework for selecting pick/topple sequences under geometric constraints.
- Published at IROS 2026.

Generative world priors for planning in partially observed rooms *2025 – 2026*

- Investigated generative models as zero-shot priors over plausible unseen environment structure for downstream robot planning.
- Generated multiple possible worlds from partial observations and evaluated planning/object-reaching decisions across the induced distribution rather than committing to a single reconstructed scene.
- Presented at RSS 2026 Workshop on Foundation Models for Robot Planning (FM4RoboPlan).

Motion planning for optimal probability of collision under environment uncertainty *Feb 2024 – Oct 2024*

- Formulated sampling-based motion planning over environment samples to optimise probability of collision-free execution rather than path length alone.
- Used roadmap search together with mixed-integer optimisation to refine candidate paths and established asymptotic optimality properties for the probabilistic objective.
- Evaluated the framework on robotic planning problems, including high-dimensional manipulation settings. Published at IROS 2024.

Arbitor: numerical emulation for DNN accelerators, University of Toronto *Sep 2022 – Jan 2023*

- Co-developed a hardware emulation tool for rapidly evaluating DNN accelerator designs and the numerical effects of common DNN optimisations.
- Leveraged machine-learning compiler infrastructure to enable accurate software emulation and fast prototyping. Published at USENIX ATC 2023.

Selected earlier research/course projects *2021 – 2023*

- Imitation learning combined with PILCO-style model-based policy learning for improved sample efficiency on higher-dimensional control tasks.
- Multi-object tracking evaluation comparing camera–LiDAR fusion and LiDAR-only baselines.
- Unsupervised text classification using dictionary learning; blind source separation using FastICA and dictionary learning;

extensions of PCA to exponential-family models.
- Near-light photometric stereo and multimodal trajectory prediction for autonomous driving.

Industry Experience

Senior Systems Engineer, Edge Processing Business Line – NXP Semiconductors Jul 2013 – Feb 2025
Freescale Semiconductor, Jul 2013 – Dec 2015; Freescale merged into NXP in 2015.

- Progressed through multiple engineering roles to Senior Systems Engineer, contributing to system solution design, embedded software development, integration, performance optimisation, technical innovation, and customer-facing deployment.
- Designed, developed, integrated, and deployed software components across embedded and edge-computing systems using C, C++, Python, Java, and architecture-specific assembly.
- Diagnosed software/hardware bottlenecks and optimised critical paths through algorithm redesign, compiler and instruction-level optimisation, cache/TLB analysis, multicore parallelism, accelerator offloading, and system-level profiling.
- Produced proof-of-concept systems, technical documentation, application notes, and engineering solutions; provided training and remote/on-site support to internal teams, partners, and customers.

Selected Technical Projects

- **Machine Learning Foundation Project:** collaborated with NXP's CTO organisation on internal infrastructure for creating, adapting, deploying, optimising, and evaluating deep-learning/foundation models for edge-AI applications.
- **NPU and Heterogeneous AI Inference Platform:** developed models and deployment workflows for NXP's Neutron AI inference engine; investigated heterogeneous execution across NPU, GPU, ARM Cortex-A, and ARM Cortex-M backends and optimisation of drivers, runtimes, firmware, and supporting software stacks.
- **i.MX 9 Smart Meter / Edge-AI Project:** developed and deployed AI algorithms on i.MX 9 NPU platforms for appliance anomaly detection, diagnosis/prediction, and non-intrusive load/energy disaggregation.
- **5G/O-RAN Systems:** contributed to 5G RU/DU, CPE, small-cell, ISC, femtocell, and O-RAN solutions across LA12xx/LS104x/LX2160-class platforms, integrating and supporting L1/L2/L3 and accelerator-enabled system components.
- **System Performance Analysis and Optimisation:** used tools such as CodeWarrior and OProfile together with compiler, architecture, cache/TLB, bus, accelerator, and multicore analysis to improve end-to-end customer software performance.

Teaching Experience

COMP3600 – Algorithms, Australian National University Jul 2026 – Present
Tutor for algorithm design and analysis, including trees, balanced search structures, randomized algorithms, graph algorithms, dynamic programming, and complexity analysis; prepare tutorial explanations, solutions, and marking guidance.

COMP8620 – Advanced Topics in Artificial Intelligence, ANU Jul 2026 – Present
Tutor for advanced AI and robotics topics including configuration spaces, motion planning, MDPs, POMDPs, belief-state planning, value/policy iteration, RTDP, and reinforcement learning.

COMP8600 – Statistical Machine Learning, ANU Feb 2026 – Jun 2026
Tutor for probabilistic modelling, Bayesian inference, kernel methods, Gaussian processes, optimisation, and statistical learning.

COMP8460 – Advanced Algorithms, ANU Feb 2026 – Jun 2026
Tutor for advanced algorithm design, randomized and approximation algorithms, graph algorithms, and complexity analysis.

COMP8620 – Advanced Topics in Artificial Intelligence, ANU Jul 2025 – Nov 2025
Delivered tutorials on motion planning, Monte Carlo methods, Bayesian inference, MDPs, POMDPs, and reinforcement learning; supported demonstrations, Q&A, and assessment.

COMP6670 – Introduction to Machine Learning, ANU Jul 2025 – Nov 2025
Tutored mathematical foundations of machine learning, including linear algebra, probability, vector calculus, optimisation, and classical ML algorithms.

C++ Programming Language Tutor Feb 2011 – Jun 2011
Delivered tutorial sessions on object-oriented programming, data structures, and debugging for undergraduate and master's students.

Guest Lecturer – Mobile Computing Feb 2012 – Jun 2012
Contributed lectures/tutorial material on mobile application development, Android, and mobile-system design.

Awards and Recognition

Academic

- ANU PhD Scholarship, 2024.
- ANU Postgraduate Medal for Academic Excellence, 2023 **(19 / ~6,000 ANU graduates)**.

Industry

- NXP Top Achiever performance rating, seven consecutive years (2015–2021/22 in historical CV records).

- NXP China R&D Innovation Award – 5G DFE Project, 2022.
- NXP Bravo Award – Major Customer LX2160 Project, 2020.
- NXP Bravo Award – Automotive LS1043 Fast Boot Enablement, 2018.
- Freescale Rock Star Award – Intelligent Network Interface Card Project, 2014.

Technical Skills

Programming: C++, Python, C, Java, MATLAB; architecture- and accelerator-specific assembly
Robotics and Planning: ROS2, MoveIt2, OMPL; motion planning, task and motion planning, kinodynamic planning, probabilistic planning, collision checking
Optimisation and Algorithms: MILP/MIQCP, Gurobi, combinatorial optimisation, graph algorithms, randomized algorithms, approximation algorithms, dynamic programming, Monte Carlo methods
Machine Learning / AI: PyTorch, TensorFlow, probabilistic modelling, generative models, model deployment and inference optimisation, edge AI, heterogeneous computing
Simulation and Systems: Isaac Sim, Gazebo, Linux/Ubuntu, Git, CMake/Make, GCC/LLVM, cross-compilation; ARM and PowerPC architectures, SIMD/NEON
Libraries / Tools: OpenCV, Boost, Eigen; profiling and low-level system performance analysis